

Received 11 June 2025, accepted 1 July 2025, date of publication 21 July 2025, date of current version 25 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3586593

RESEARCH ARTICLE

Integration of MATH41 and Generative AI in Pre-Service Mathematics Teacher Education: An Empirical Study on Lesson Design Competency

SEJUN OH^{ID}, (Member, IEEE)

Department of Mathematics Education, Hongik University, Seoul 04066, South Korea

e-mail: soh@hongik.ac.kr

This work was supported by the 2025 Hongik University Research Fund.

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Hongik University Institutional Review Board (IRB) under Application No. 7002340-202411-HR-029, and performed in line with the Declaration of Helsinki.

ABSTRACT Generative artificial intelligence (AI) tools are rapidly transforming mathematics education by enabling automated problem generation, dynamic visualizations, and adaptive learning experiences. This study presents an empirical study on incorporating MATH41 into a 15-week pre-service teacher preparation course for mathematics majors. Twenty-four participants learned to create parameterized math problems, generate vector graphics. Following guided training, each participant designed and micro-taught a 20-minute lesson incorporating MATH41-generated resources. Reflection journals and final lesson plans were analyzed thematically, while 20 participants completed a self-assessment survey on lesson design competency. Results reveal that the automated problem generation capabilities motivated pre-service teachers to explore a broader range of instructional strategies, including personalized tasks and diverse problem variants. Reflection data indicate that while integrating AI tools can significantly boost confidence and creativity in lesson planning, careful pedagogical alignment remains essential to avoid superficial learning. Participants underscored the importance of maintaining teacher oversight—especially when adapting AI-generated problems for particular learner needs. Additionally, their post-course self-assessments showed high confidence in digital tool integration, yet they acknowledged that anticipating student misconceptions requires further field experience. Overall, this study contributes to understanding how teacher education programs can enhance lesson design competencies via structured AI tool integration. It also highlights the critical role of reflective practice in ensuring that automated content creation fosters deeper instructional effectiveness rather than uncritical AI dependency.

INDEX TERMS Generative AI, mathematics education, pre-service teachers, MATH41, problem generation, instructional design competency, teacher reflection, AI in education, lesson planning.

I. INTRODUCTION

Recent advances in generative artificial intelligence (AI), such as GPT-based chatbots and automatic problem generation tools, are reshaping the landscape of education. In mathematics education, these tools can generate realistic and contextualized math problems, offering new ways to engage students in problem-solving. For example, large language models like ChatGPT can rapidly produce custom

math tasks, potentially freeing teachers from time-consuming problem creation and allowing more focus on facilitation. Specialized platforms like *MATH41* extend this capability by enabling teachers to algorithmically generate practice problems and even graphical illustrations using TeX and Python code. With such technology, teachers could provide students with *personalized and challenging tasks* at scale. Given these possibilities, it is crucial that future teachers are prepared to integrate AI tools into their lesson design and teaching practices [1], [2], [3], [4], [5], [6]. Integrating AI tools in education is not without challenges. Prior research indicates that when

The associate editor coordinating the review of this manuscript and approving it for publication was Alba Amato^{ID}.

used thoughtfully, AI assistance can enhance learning outcomes. For example, intelligent tutoring systems have generally shown positive effects on K–12 student performance [1], [2], [7], [8]. Generative AI tools such as ChatGPT can also support mathematics instruction by generating realistic, contextualized problems and adaptive practice tasks. This helps teachers foster students' *number sense*, engage them in rich problem-solving activities, and differentiate instruction to individual learning needs [1], [2], [8], [9]. These capabilities align with research-based strategies in math education and illustrate AI's pedagogical potential in the classroom.

However, if these tools are used without proper pedagogical context or guidance, they may hinder deep learning and skill acquisition [1], [2], [3], [10]. Some students who had leaned on AI to supply answers struggled to solve problems independently afterwards [1], [2]. This outcome highlights the risk of over-reliance on AI, where short-term gains come at the expense of long-term understanding. Recent reviews of AI in education likewise caution that unrestricted or unguided use of AI can undermine learning, emphasizing the importance of thoughtful integration and teacher oversight [1], [2], [3], [4], [5], [6], [9], [11]. By providing appropriate scaffolding and ensuring students remain actively engaged in problem-solving, educators can harness AI as a powerful tool without sacrificing deep conceptual learning.

In light of these developments, this study explores the incorporation of a generative AI-driven tool in pre-service teacher education for mathematics. While emerging work has begun examining pre-service teachers' use of tools like ChatGPT for lesson planning, there is a need to design and evaluate structured programs that prepare teachers to use such tools effectively. This study focuses on *MATH41*, an automatic problem generation and visualization system. Problem-solving competency and lesson design competency are key skills for mathematics teachers, and this study hypothesizes that guided use of tools like MATH41 can enhance these areas. Building on theories of reflection-based teacher learning and technology integration (e.g., TPACK), this study designed a 15-week pre-service teacher training program embedding tool use into lesson planning and instruction. This paper reports on the design of the program and analyzes its outcomes in terms of the participants' lesson design abilities, perceptions of AI tools in education, and reflective insights. The findings contribute to understanding how AI tool integration in teacher education can strengthen future educators' competencies and readiness for technology-enhanced teaching.

II. THEORETICAL BACKGROUND

A. GENERATIVE AI AND AUTOMATIC PROBLEM GENERATION IN MATH EDUCATION

Generative AI tools and automatic problem generators are increasingly used in mathematics education to provide personalized and adaptive learning experiences [8], [12], [13]. Research indicates that AI-based systems can enhance student performance in math by tailoring content

and feedback to individual needs [2], [12]. For example, a recent study used a GPT-based system to generate contextualized math word problems for 5th-grade geometry; students who learned with these AI-personalized problems showed significantly better learning outcomes than those without such personalization [10]. AI can also dynamically adjust problem contexts to align with student interests and interpret students' solution attempts [7]. In practice, many teachers have responded optimistically to these tools: in one survey, K–12 teachers expressed positive attitudes and a willingness to integrate generative AI into their teaching [2], [4], noting that it can save time and automate routine tasks in lesson planning and materials creation.

However, the effectiveness of generative AI in math education depends on how it is used. Over-reliance on AI without guidance may hinder deep learning. For instance, an experiment with high school math students found that those allowed to use an unrestricted AI (ChatGPT-like) assistant during practice scored 48% higher in practice sessions than a control group, but when tested later without AI, they performed 17% worse than the control, suggesting they hadn't fully learned the material [14]. By contrast, another group of students who practiced with a guided AI tutor (which provided hints instead of direct answers) achieved large practice gains (127% better during practice) while still performing on par with the control on the post-test [14]. These results highlight the importance of proper scaffolding and “guardrails” when using AI: AI can accelerate problem-solving and practice, but educators must ensure students engage cognitively with the content rather than simply outsourcing their thinking.

Recent studies also explore *how* to harness generative AI effectively for math problem generation. Advanced large language models (LLMs) like GPT-4 can produce high-quality math questions and step-by-step solutions when given appropriate prompts [2], [13]. For example, [2] found that with carefully engineered prompts (e.g. chain-of-thought reasoning), GPT-3.5 and GPT-4 could automatically generate valid STEM assessment items in mathematics (and science), as validated by experts. This suggests automated problem generators can reliably assist teachers in creating practice exercises or quizzes at scale. Nevertheless, researchers caution that AI-generated content must be vetted for accuracy and alignment with learning goals, since LLMs can sometimes “hallucinate” or introduce misconceptions [2]. Overall, the literature indicates that generative AI, used thoughtfully, offers promising benefits in math education – from improving student achievement through personalized practice to increasing teachers' efficiency in developing instructional materials – while also underscoring the need for guidance and ethical use to avoid pitfalls.

B. AI IN PRE-SERVICE TEACHER EDUCATION FOR LESSON DESIGN AND PROBLEM-SOLVING

The rapid emergence of AI in education is also transforming pre-service teacher education, particularly in how pre-service teachers learn to design lessons and develop problem-solving

pedagogy. Teacher education programs today are “*compelled to prepare pre-service teachers with the necessary skills to integrate these advanced technologies into their daily teaching practices*” [4], [10], [15]. This includes building competencies in using AI tools for lesson planning, instructional design, and data-driven decision making. The consensus is that future teachers need to develop Technological, Pedagogical, and Content Knowledge (TPACK) so they can leverage AI effectively while maintaining sound pedagogy. Recent studies have begun documenting cases of pre-service teachers using AI and the outcomes on their lesson design abilities.

One qualitative study examined how elementary pre-service teachers used ChatGPT as a planning assistant when creating unit plans [10]. The researchers found that pre-service teachers engaged in “*complex and nuanced interactions*” with the AI and could use it to generate ideas and draft content, but successful use required them to draw upon their pedagogical and content knowledge to vet and adapt the AI’s suggestions. Concrete case studies illustrate both the benefits and challenges of using AI in pre-service teacher lesson design [1], [3]. In a recent “*Science Methods*” course, 20 pre-service teachers used Microsoft’s Copilot (an AI assistant similar to ChatGPT) to help plan 4th-grade science lessons [3]. Analysis of their reflection journals and focus group discussions revealed that the pre-service teachers primarily used the AI to generate lesson plan outlines, teaching materials (e.g. visuals, examples), assessment questions, and even suggestions for instructional strategies. The pre-service teachers reported several clear benefits of the AI: it sparked new ideas and approaches they hadn’t considered, helped attract attention to overlooked aspects of their lessons, and significantly saved time and effort in drafting plans and resources. They also found the AI useful for tailoring content to students’ needs (e.g. adjusting difficulty, providing analogies), which suggests it can enhance differentiation. On the other hand, the pre-service teachers noted some concerns – the AI’s involvement meant less human interaction in the planning process (e.g. fewer discussions with peers) and raised the risk of dependency on the tool. These reflections underscore that while AI can empower novice teachers by extending their capabilities, it should be used to augment rather than replace traditional lesson design practices.

Similarly, a case study of Korean pre-service English teachers using generative AI for lesson planning found that AI tools facilitated efficiency and creativity in their planning process [4]. The pre-service teachers used AI for tasks like choosing lesson topics, creating example activities/materials, structuring the lesson flow, and even checking grammar or phrasing in their plans. They reported that the AI helped them generate content more quickly and inspired more creative lesson ideas. However, notable challenges emerged: the quality of AI outputs varied, sometimes requiring substantial editing, and the AI-suggested lessons often lacked robust pedagogical strategies (for instance, activities were superficially engaging but not deep in inquiry). This aligns with other findings that

AI can propose *initial* plans which then need the teacher’s pedagogical refinement [3], [9], [16]. Kerr and Kim [4], who conducted the study, conclude that targeted training is needed in “*prompt engineering, ethical considerations, pedagogy, and collaborative practices*” to fully enhance pre-service teachers’ AI competencies for lesson design. In other words, teacher preparation should explicitly teach future teachers how to query AI tools effectively, scrutinize and adapt the AI’s suggestions, and consider ethical issues (e.g. bias, academic integrity) when using AI in education.

Across these cases, a common theme is that AI can act as a creative partner for pre-service teachers, expanding their repertoire of ideas and helping them solve lesson-planning problems (like finding suitable examples or alternative explanations). Pre-service teachers often noted that AI assistance gave them “*new ideas beyond [their] existing knowledge*” and brought up aspects “*that never crossed their minds,*” as well as provided quick feedback on their plans [1], [3]. This suggests AI tools can help novice teachers overcome limited experience by exposing them to a wider variety of approaches. Moreover, AI can handle some time-consuming tasks (e.g. generating quiz questions, formatting lesson plan templates), allowing pre-service teachers to focus more on high-level design and student thinking. Crucially, though, using AI in teacher education also serves as a learning opportunity: by critically analyzing AI outputs, pre-service teachers practice higher-order thinking about pedagogy (why a given AI-suggested activity may or may not work). Several educators have pointed out that engaging pre-service teachers with AI in a reflective manner can strengthen their understanding of content and pedagogy – essentially, the pre-service teachers learn by “teaching” or correcting the AI. In sum, early evidence from pre-service teacher programs indicates that AI integration can enhance pre-service teachers’ lesson design and problem-solving skills, if teacher educators guide pre-service teachers to use these tools thoughtfully. The experiences highlight the need for developing digital pedagogical skills (TPACK) alongside a critical mindset, so that new teachers enter the profession ready to leverage AI for innovative lesson design while upholding strong pedagogical principles [1], [3], [4], [10], [15], [17], [18], [19].

C. REFLECTION-BASED EDUCATIONAL ASSESSMENT

Reflective practice is a long-standing pillar of teacher education and is increasingly used as an assessment strategy in education. “Reflection-based assessment” typically involves having learners (whether student teachers or K-12 students) articulate their experiences, thought processes, and growth in writing (e.g. journals, portfolios) which are then analyzed to evaluate learning. Such reflective journals or diaries serve a dual purpose: they prompt learners to engage in metacognition and self-assessment, and they provide instructors with rich qualitative data on the learner’s development that traditional tests might miss [20], [21], [22]. Prior research has found that reflective writing tasks can function as effective

formative assessment tools, helping students identify gaps in understanding and regulate their own learning. For example, having students keep reflective diaries was shown to help them become more self-regulated learners, as they set goals, monitored their progress, and adjusted strategies based on their reflections [23]. In this way, reflection-based assessments align with the goals of *assessment for learning* – they not only evaluate learning after the fact but actively contribute to learning by encouraging introspection and continuous improvement.

In teacher preparation specifically, reflective journals are often used to assess professional growth. Pre-service teachers are typically asked to reflect on their classroom experiences, lesson implementations, or micro-teaching sessions. These reflections can be analyzed by instructors or researchers to gauge the development of teaching competencies, belief changes, and readiness for the classroom. For instance, in a longitudinal elective course where medical students were trained in teaching skills, students kept bi-weekly reflective teaching journals. Analysis of these journals revealed key themes of growth: the reflections provided “*a framework for pedagogical skill building*,” acted as a “*sense-making tool*” for the complex social environment of the classroom and facilitated “*educator identity formation*” [21]. Over just a few months, the student-teachers’ self-reported confidence in teaching climbed markedly, which the authors attributed in part to the structured reflection and feedback cycle built into the journal process. This example underscores how embedded reflection assessments can document and foster growth in teaching skills and confidence. The journals made the students’ thought processes and struggles visible, allowing both the students themselves and their mentors to identify areas for improvement and adjust teaching strategies accordingly (e.g. overcoming “imposter syndrome” as noted in the reflections, or improving specific pedagogical techniques).

There are also numerous cases of using reflection analysis as an assessment in general education. In language learning, for example, having ESL students write reflective diaries about their learning process has been used as a formative assessment strategy to improve meta-cognitive and motivational aspects of learning. One study in a Gulf context found that reflective diary writing, coupled with teacher feedback, led to improved self-regulation and motivation among middle school ESL students [20]. Students began to take more ownership of their learning, adjusting their study habits based on insights from their own reflections and the feedback received. In higher education, reflective essays or portfolios are sometimes used as a summative assessment of learning in capstone courses or internships, on the premise that the ability to thoughtfully reflect on one’s experience is evidence of deeper learning. A review of practices in professional degree programs found that reflective writing is commonly incorporated into evaluation, especially in fields like nursing and education, to assess not just factual knowledge but also

the student’s ability to integrate theory with practice and to engage in lifelong learning [12], [20], [21], [22], [24].

From a methodological standpoint, educators and researchers have developed rubrics and analytic techniques to evaluate the content of reflection journals systematically. This is important to ensure that reflection-based assessment is reliable and valid (i.e. not just anecdotal). For example, one recent study demonstrated a mixed-method approach to evaluating pre-service teachers’ reflective writing: the authors combined qualitative content analysis with computational text analysis (using NLP tools and topic modeling) to categorize reflections by level and focus [12], [22]. They showed that it is feasible to assess reflections across multiple dimensions – cognitive (e.g. evidence of critical thinking), affective (emotions, attitudes expressed), and thematic (the topics or issues discussed) – which can then be correlated with other measures of performance [22]. Such approaches allow reflection journals to be scored or profiled in a more objective manner, turning qualitative reflections into actionable assessment data. For instance, a teacher educator might rate a student teacher’s journal entries on a reflection depth scale (descriptive, analytical, critical, etc.), and use that to monitor progress over time or to identify if further coaching is needed. In research, content analysis of reflective journals has been used to compare different interventions: one study found that prompting pre-service teachers with specific guiding questions in their journals elicited reflective responses [9], [21], [24], [25]. Reflection-based assessment is a versatile and powerful practice – it not only provides evidence of learning (often capturing outcomes like professional identity, self-awareness, or problem-solving skills that traditional tests miss), but the very act of reflection benefits learners. The literature supports using structured reflection (with clear prompts, rubrics, and feedback) as part of the assessment strategy to promote deeper learning and self-improvement.

D. LINKING LESSON DESIGN COMPETENCY AND DIGITAL TOOL USE

In modern education, a teacher’s lesson design competency is increasingly intertwined with their ability to effectively use digital tools. As classrooms become more technology-rich, teachers are expected to integrate digital resources, educational software, and now AI tools into their lesson plans to enhance learning. Theoretical frameworks like TPACK (Technological Pedagogical Content Knowledge) encapsulate this integration, arguing that expert teachers skillfully blend content knowledge, pedagogy, and technology when designing instruction. In other words, knowing *how* to use digital tools is now a part of knowing *how* to design a good lesson [1], [4], [5], [26]. Teacher educators emphasize that today’s teachers must develop “professional digital competence” alongside traditional pedagogical skills [4], [27]. A recent review echoes this, noting broad agreement that lesson planning is the critical stage for technology integration

in the classroom – it’s during planning that teachers decide *if* and *how* to embed digital tools in activities to meet learning objectives [4]. Simply put, effective lesson design in the 21st century inherently involves thoughtful use of technology.

Empirical studies support a strong connection between digital tool use and lesson design quality. Teachers who are more proficient with digital technologies tend to create more innovative, student-centered lessons that incorporate those tools, and vice versa, engaging in tech integration can improve teachers’ instructional design skills [2], [4], [5], [11], [26]. For instance, a study on pre-service teachers developed a diagnostic scale for digital integration competency and explicitly included lesson planning as a key dimension of that competency. The scale assesses pre-service teachers’ self-efficacy in integrating digital technologies in lesson planning, implementation, and reflection phases of teaching [28]. This underscores that the field recognizes “using digital tools in lesson design” as a measurable skill set. Other research has found that professional development focused on digital lesson design can significantly enhance teachers’ capabilities. In one case, in-service teachers who participated in a collaborative workshop to incorporate digital media into lesson study not only gained confidence in using those tools but also developed a structured lesson design checklist for technology integration (i.e. a guide to deciding when and why to use a digital resource in a lesson) [29]. This kind of outcome illustrates how improving teachers’ digital tool usage goes hand-in-hand with improving their lesson planning competence.

Moreover, teachers’ attitudes and acceptance of technology influence their design practices. Studies show that teachers with higher digital competence are more likely to embrace new tools and integrate them into their planning. For example, one survey study found a positive correlation between teachers’ self-rated digital competence and their intention to design technology-enhanced lessons (those confident in their tech skills more readily planned to use digital apps for student collaboration, assessment, etc.) [26], [27], [28]. On the flip side, teachers who lack digital self-efficacy tend to stick to traditional lesson formats, which can limit opportunities for students to engage with multimedia content or adaptive learning systems. Thus, building teachers’ comfort and skill with digital tools is often a prerequisite to seeing those tools used meaningfully in lesson plans.

It is also notable that policy and standards frameworks now explicitly link lesson design and digital competence. For instance, the European framework DigCompEdu outlines detailed competencies for teachers, including the ability to “select, create and modify digital educational resources” and “plan for and implement digital devices and software in the teaching process”. Many teacher education programs worldwide have adopted similar competency models, integrating technology training into lesson planning courses [3], [4], [15], [30]. The OECD, in a 2023 report on teacher digital competence, highlighted that formal criteria for teacher quality in

some countries now include effective use of ICT in planning and instruction [5]. All this reflects a paradigm shift: digital tool usage is not an optional add-on, but a core component of lesson design competence in contemporary education.

The literature and prevailing educational standards suggest a tight coupling between lesson design skills and digital tool utilization. Teachers who effectively use digital tools (from interactive simulations in math to AI-driven feedback systems) typically design richer and more adaptive learning experiences. Conversely, strong lesson design capacity enables teachers to integrate technology in a pedagogically sound way (rather than using tech for tech’s sake). Therefore, developing teachers’ digital competencies – including emerging areas like AI literacy – is viewed as essential to enhancing their overall instructional design ability. As one review put it, lesson planning is the crucible for technology integration [26]– it’s where the synergy of pedagogy and technology is realized. For researchers and practitioners, this means initiatives aimed at improving lesson design (e.g. lesson study, design-based workshops) should incorporate training on digital tools, and vice versa, ICT training for teachers should be anchored in authentic lesson planning tasks. This integrated approach ensures that teachers can confidently harness digital innovations to design effective, engaging lessons that meet the needs of today’s learners.

III. METHODOLOGY

A. PARTICIPANTS AND CONTEXT

The study was conducted with 24 pre-service mathematics teachers enrolled in a teaching methods course at a university (two class sections, total $N = 24$). These participants were second-year students in the teacher education program. The course was a 15-week program where, in addition to regular pedagogy content, an math instruction tool (MATH41) was introduced and integrated into course activities. All participants gave informed consent for their data to be used in this research.

B. INSTRUCTIONAL DESIGN AND PROCEDURE

Over the 15-week semester, the course incorporated the use of MATH41 into a cycle of lesson design, teaching practice, and reflection. Table 1 summarizes the weekly course sequence and core activities implemented throughout the 15-week program.

The weekly structure was as follows:

- **Week 1: Orientation** – Introduction to the course objectives, overview of AI in education, and pre-service teachers’ prior experiences with technology were discussed. The importance of integrating digital tools in lesson design was emphasized to set the stage.
- **Weeks 2–4: Training in MATH41 Usage** – Participants received hands-on training on MATH41, an online platform for generating math problems and visuals. They practiced using the MATH41 editor to create custom math problems using TeX (for mathematical formatting)

TABLE 1. Weekly course sequence and activities.

Week (s)	Activity	Summary
Week 1	Orientation	Introduce course objectives, overview AI in education, discuss prior tech experiences, and emphasize digital tool integration in lesson design.
Weeks 2–4	MATH41 Training & Prompt Engineering	Hands-on practice with TeX and Python for problem generation, use Data Canvas for visuals, and host a prompt-engineering workshop to refine AI queries.
Weeks 5–10	Lesson Planning with AI Support	Design comprehensive lesson plans using MATH41-generated problems and visuals, iteratively refine with peer/instructor feedback, and ensure pedagogical alignment.
Weeks 11–12	Teaching Demonstrations (Microteaching)	Conduct and video-record 20-minute microteaching sessions using AI-created materials to simulate real classroom instruction.
Weeks 13–14	Reflection & Revision	Write detailed reflection journals, review feedback from peer pre-service teachers, and revise lesson plans based on insights.
Week 15	Post-Survey & Discussion	Administer self-assessment survey on lesson design competency, hold class debrief, and summarize key learnings and implications.

and simple Python scripts (for algorithmic problem generation). They also learned to use MATH41’s *Data Canvas* feature to produce graphs or geometric figures from code. This practice allowed them to become comfortable with the tool’s capabilities, such as inserting randomized variables into problem templates and auto-generating solution graphics.

- **Weeks 5–10: Lesson Planning with AI Support** – Each pre-service teacher was tasked with designing a comprehensive lesson plan for a secondary math topic (of their choice or assigned) while integrating MATH41. They formulated learning objectives and then used MATH41 to generate practice problems, examples, or visual aids aligned with those objectives. During this period, they iteratively refined their lesson plans. Peers and instructors provided feedback, and participants were encouraged to consider how the AI-generated content would be used pedagogically (not just as a convenience).

The focus was on maintaining pedagogical alignment – ensuring the automatically generated problems or visuals served the learning goals and matched the students’ level.

Notably, this study provided guidance on prompt engineering to the participants. In Week 2, a brief training session illustrated how the phrasing of a query significantly shapes AI outputs. For instance, a generic prompt like ‘Create a math lesson plan’ yielded only broad suggestions, whereas a specific prompt — ‘Generate a 45-minute 7th-grade math lesson on the Pythagorean Theorem with a hands-on activity’ — produced a far more detailed and relevant plan. Throughout the course, candidates learned to iteratively refine their prompts. Many initially struggled to get useful answers due to poorly worded questions, echoing findings that inadequate prompting hampers AI utility [1], [2], [6]. By practicing and sharing strategies, they learned to script prompts effectively, which elicited more accurate and context-appropriate responses. This study observed that as prompt quality improved, the AI’s contributions became more aligned with curricular goals and student needs. This emphasis on prompt technique not only improved lesson outputs but also fostered an understanding that mastery of questioning is vital when using AI in pedagogy.

- **Weeks 11–12: Teaching Demonstrations (Microteaching)** – Participants conducted a 20-minute segment of their designed lesson in a microteaching session. In these sessions, each pre-service teacher taught their lesson excerpt to their peers (who role-played as students). They utilized the materials created with MATH41 during instruction – for example, presenting problems or graphs generated by the tool. These sessions were video recorded for subsequent analysis.
- **Weeks 13–14: Reflection and Revision** – After teaching, each participant wrote a detailed reflection journal about their lesson planning and teaching experience. Prompts in the reflection asked them to evaluate the effectiveness of their lesson design, how the AI tool and MATH41 contributed to or altered their approach, challenges faced, and what they would improve next time. Participants were encouraged to connect theory to practice, commenting on how using MATH41 influenced student engagement or understanding in their microteaching. In addition to the instructor, all observing pre-service teachers provided feedback on their peers’ lessons.
- **Week 15: Post-survey and Discussion** – In the final week, a post-course survey was administered to assess the participants’ self-evaluated lesson design competency after the program. This survey (described below) allowed the teachers to reflect quantitatively on their skills. A whole class debrief discussion followed, where common experiences were shared. The course concluded with a summary of learnings and implications for the pre-service teachers’ future classrooms.

C. DATA COLLECTION

Multiple data sources were collected to evaluate the outcomes of the program:

- **Self-Assessment Survey:** A post-program questionnaire on *lesson design competency* was administered to the pre-service teachers. Twenty out of 24 participants responded (response rate: 83%). The survey prompted pre-service teachers to rate their agreement (on a 4-point Likert scale) with statements regarding their ability to design effective math lessons. It covered facets such as aligning lessons with objectives, incorporating appropriate problems and examples, integrating technology tools, and adapting designs to student needs. This survey, being a self-report measure, captured the participants' confidence and perceived growth in lesson planning skills after experiencing AI-integrated training.
- **Lesson Plan Artifacts:** Each participant submitted their final written lesson plan (including any worksheets or materials generated with MATH41). These documents illustrate how MATH41 was embedded (e.g., screenshots of problems, graphs from the Data Canvas). They provide evidence of the way pre-service teachers used the tool in their planning.
- **Teaching Demonstration Videos:** The video recordings of the 20-minute microteaching sessions were collected. These were later reviewed to observe how the lesson plan translated to teaching practice, including how AI-generated content was introduced to "students." Due to time constraints, a detailed analysis of the videos is outside the scope of this paper; however, they were used to contextualize reflection comments and were reviewed by the external teachers for an overall impression of teaching effectiveness. Specifically, quantitative or qualitative in-depth analysis of the videos was not performed. Instead, specific video content was informally reviewed and cross-referenced with lesson plans to understand participants' reflections and ensure the credibility of self-reported data.
- **Reflection Journals:** The written reflections (one per participant) were perhaps the richest qualitative data. In these journals, pre-service teachers often candidly described their thought processes and reactions to using MATH41. Two veteran math teachers acted as evaluators, reading reflection journals and Lesson Plan Artifacts to identify recurring themes and notable insights (what this study term "exemplary reflection cases"). Each reflection was labeled anonymously (S1 through S24) to protect identities. The external reviewers' feedback on these reflections was also recorded, which helped triangulate the findings and ensure that the researchers' interpretations aligned with an experienced educator's perspective.

D. ETHICAL USE OF AI

The course included explicit guidelines on the responsible use of generative AI. Pre-service teachers were cautioned about issues of academic integrity – for instance, they were instructed that copying AI-generated text verbatim in assignments constitutes plagiarism, a misuse that was strictly prohibited. Pre-service teachers were instead encouraged to use AI suggestions as inspiration and to cite any AI-derived content as appropriate. This study also addressed potential biases in AI outputs: participants learned to critically evaluate the AI's responses for any social or cultural bias, aligning with ethical AI usage practice. Moreover, data privacy considerations were discussed; students refrained from inputting sensitive personal data into the AI system. Overall, human oversight was maintained at every stage – AI-generated materials were reviewed and vetted by the pre-service teachers (and instructors) to ensure accuracy, fairness, and alignment with educational values.

E. DATA ANALYSIS

This study employed a mixed-methods analysis approach. The quantitative survey data from the 20 respondents were analyzed for central tendencies (mean and standard deviation for each competency item). Since the survey was only administered post-intervention (without a pre-survey), this study interpreted the results in light of participants' qualitative comments about how their skills changed, rather than conducting statistical comparisons. A summary table was constructed to report key competency areas and average self-ratings (see Results section).

The qualitative data (reflection journals) were analyzed using thematic analysis by researcher who conducted repeated analyses over time to identify emergent themes related to (a) perceptions of AI tool usefulness or drawbacks, (b) changes in lesson design approach, (c) problem-solving pedagogy insights, and (d) remaining challenges or concerns. Two external in-service teacher experts reviewed and validated the coding, highlighting particular reflections that demonstrated significant growth or critical thinking. Representative quotes were then extracted to illustrate each theme. The iterative analysis by the researcher, together with the expert validation, ensured a robust and reliable coding process. Any discrepancies between the researcher's analysis and the external experts' observations were discussed until a consensus was reached.

The lesson plans and videos were reviewed informally to contextualize the above analyses. For instance, if a participant wrote about how MATH41 enabled them to include more diagrams, this study verified this by checking their lesson materials. This helped ensure the credibility of self-reported data. Overall, by triangulating survey results with reflections and artifacts, this study aimed to develop a well-rounded understanding of the program's impact on the pre-service teachers.

IV. RESULTS

A. MATH41 SYSTEM FUNCTIONALITY AND USAGE IN LESSON DESIGN

MATH41 is a cloud-based authoring environment that unifies (i) LaTeX-compatible TeX typesetting for precise mathematical notation, (ii) an embedded Python engine for algorithmic problem generation, and (iii) a vector-graphics “Data Canvas” for dynamic diagramming. Because the platform runs entirely in a web browser, teachers—even those with minimal coding background—can script parameterized tasks, preview fully rendered output in real time, and export ready-to-use materials without additional software. Participants particularly highlighted specific benefits of these features for their lesson design:

- **Efficiency and Variation from Parameterized Problem Generation:** Teachers can define a problem template in TeX, insert variables or parameters, and attach a Python snippet to generate values or compute answers. Participants emphasized this capability for swiftly producing diverse and customized problem sets, significantly reducing repetitive tasks and freeing up cognitive resources for creative and pedagogical decision-making.

Figure 1 displays the interface of the MATH41 editor, which is organized into three main columns. The left column contains curriculum tags and problem classification settings, allowing users to align generated problems with specific educational standards or competencies. The middle column includes the problem, answer, explanation, and multiple-choice options written in TeX, along with the corresponding Python code below. This Python code defines variables—such as coefficients, roots, or ranges—that are randomized each time the problem is rendered. Participants identified this Python-based randomization as particularly beneficial for supporting differentiated instruction, as it allowed easy adaptation of problem complexity and context according to learner needs.

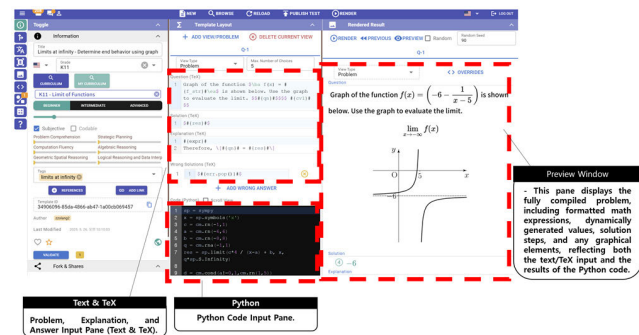


FIGURE 1. Example of the MATH41 editor interface showing integration of problem content and python-based randomization.

The right column shows the fully rendered output, where mathematical expressions, tables, and graphs are dynamically compiled and visualized. This layout allows pre-service teachers to efficiently create a wide variety of practice problems and visual aids by modifying a single problem template. The seamless integration of TeX and Python empowers users

to combine precise formatting with algorithmic variation, supporting both scalability and pedagogical alignment in lesson planning.

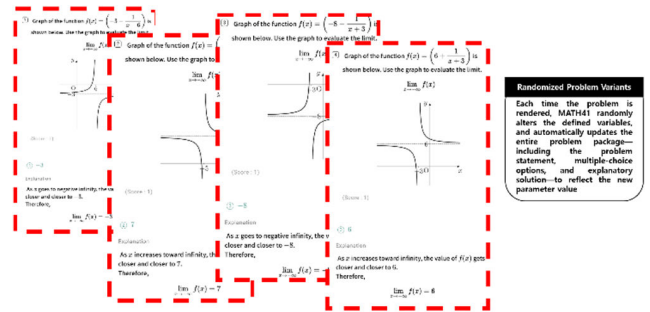


FIGURE 2. This figure displays four automatically generated instances of the same math problem, created using the MATH41 platform.

The auto-generation functionality of MATH41 is exemplified in Figure 2, which shows four distinct renderings of a limit problem involving rational functions. The original task asks students to evaluate the limit of a given rational function as x approaches a specific value (either a constant or infinity). This requires analyzing the behavior of the function as x approaches the limit point and interpreting its graph.

As seen in the figure, MATH41 allows teachers to randomly alter the parameters within the rational function (e.g., coefficients, constants, and the limit point). Each rendered version of the problem maintains the same mathematical structure and logic but differs in specific numerical values. The corresponding solutions, explanations, and graphs are recalculated and displayed automatically for each instance. Each time the problem is rendered, MATH41 randomly alters the defined variables and automatically updates the entire problem package—including the problem statement, multiple-choice options, and explanatory solution—to reflect the new parameter value. This randomization capability was repeatedly praised by participants for facilitating differentiation and repetitive practice with conceptual consistency, which they recognized as crucial for addressing student misconceptions and reinforcing deeper understanding.

- **Visual Clarity via Data Canvas:** Another strength is the Data Canvas tool within MATH41, which enables automatic creation of vector graphics. Teachers can choose from pre-defined shapes (points, lines, circles, etc.) or graph functions by writing simple code. The Data Canvas will then render a high-quality graph or diagram that can be embedded in the problem or exported. Participants strongly appreciated this functionality as it substantially lowered the barrier to including visually clear and accurate diagrams, enhancing their ability to clearly communicate mathematical concepts. Figure 3 illustrates the Data Canvas interface that the participants used. On the left, a menu of shapes and plot options is available (e.g., points, lines, polygons, function plots), which the user can select and configure. On the right,

the system generates the corresponding graph based on the defined elements or equations.

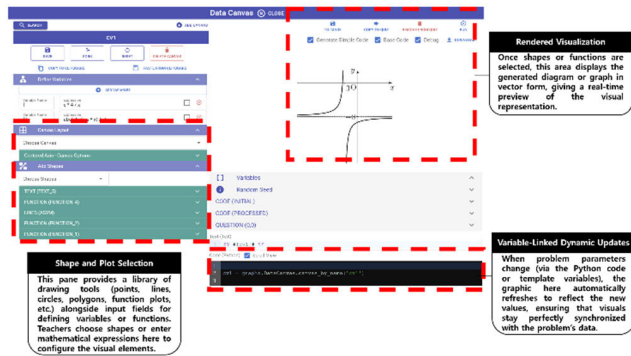


FIGURE 3. MATH41's data canvas interface, which allows automatic generation of graphs and diagrams from code. The left panel shows tools to define variables and shapes, and the right panel displays the generated graph. This feature enables teachers to create custom visuals that are perfectly aligned with the problem parameters.

The example shown is a rational function graph automatically drawn by the system after the user inputs the function and defines its parameters or equations. Participants specifically highlighted the ease and quality of visuals produced through Data Canvas, which otherwise would have required significantly more time or graphical expertise.

All 24 pre-service teachers successfully learned to use these features of MATH41 during the training phase. In their lesson plans, every participant incorporated at least one element created via MATH41. A common use-case was generating practice problem sets: for example, a participant designing a lesson on quadratic equations used Python in MATH41 to generate a set of quadratic equations with randomized coefficients for students to factor, ensuring each student might get a unique equation. Another participant focused on geometry employed the Data Canvas to produce a series of transformed geometric figures (triangles demonstrating congruence and similarity) as part of an inquiry task – something that would have been tedious to draw by hand. These artifacts demonstrate how MATH41 served as a *labor-saving device* (generating multiple problem instances with ease) and as a *creative aid* (producing visuals that might inspire new ways to present content).

From the system functionality analysis, this study observed that pre-service teachers leveraged MATH41 primarily in two ways: (a) to increase the *variety and volume of practice problems* available in their lesson, and (b) to create *dynamic visual aids* to support concept understanding. The first addresses the problem of limited examples – with automatic generation, teachers can give students additional problems of similar structure for reinforcement or adapt difficulty on the fly. Participants underscored how this feature significantly enhanced their ability to implement differentiated and personalized instruction. The second speaks to the use of visual representations in math teaching – Data Canvas lowered the barrier to including graphs/diagrams, so even less artistically inclined teachers could provide accurate drawings (for

instance, plotting a trigonometric function or illustrating an angle in standard position). Participants repeatedly noted the increased confidence and efficiency this brought to their lesson planning. These align with the study's goal of enhancing problem-solving opportunities and lesson design quality. As one participant noted in class, “*I usually would avoid including too many graphs because it's hard to make them, but now I can generate any graph I need quickly.*” In summary, the rich functionality of MATH41 was well-utilized by the teachers in designing their lessons, giving them greater flexibility and efficiency in creating instructional materials. As explicitly confirmed through participants' reflective feedback.

B. EDUCATIONAL OUTCOMES: LESSON DESIGN COMPETENCY AND REFLECTIONS

To evaluate the educational impact of integrating the AI tool, this study examined both the survey results of lesson design competency and the qualitative insights from reflection journals.

1) SELF-ASSESSMENT OF LESSON DESIGN COMPETENCY

Table 2 summarizes the post-program survey results (N = 20 respondents). Participants rated various aspects of their lesson planning ability after undergoing the AI-integrated training. Ratings were on a scale of 1 (strongly disagree) to 4 (strongly agree). Overall, the self-assessments indicate moderate confidence, with mean scores ranging from 2.90 to 3.30, rather than above 3 as previously anticipated. Notably, the item “I can provide more accurate answers and explanations in response to students' questions” received the highest mean score of **3.30**, suggesting that the participants felt relatively more confident in addressing student inquiries accurately. In contrast, the item “I know how to engage students' interest in mathematics” obtained the lowest mean score of **2.90**, implying that strategies for stimulating student engagement may require further development. Additionally, items such as “I will strive to find better ways to teach mathematics using technology in the future” (mean = **3.20**) and “I believe that, as a future mathematics teacher, I will be able to deliver lessons that are superior to those of other teachers” (mean = **3.10**) reflect a moderate level of self-assessed competency. The standard deviations, ranging from **0.44 to 0.77**, indicate moderate agreement among participants.

These survey results suggest that the program was effective in building the pre-service teachers' confidence in lesson planning, including the integration of AI tools. While this study lacks a direct pre-program baseline measure, many participants mentioned informally (in reflections and class discussion) that *initially they were unsure how to use such tools in teaching*. By the end, their high self-rate in technology integration. By the end of the program, the moderate self-ratings in technology integration hint at a gradual but incomplete shift in their comfort level. These findings align with recent research indicating that, although proper training can enhance pre-service teachers' AI-related technological

TABLE 2. Self-assessed lesson design competencies after the program (N = 20).

Survey Item	Mean	Std. Dev.
1. I will strive to find better ways to teach mathematics using technology in the future.	3.2	0.77
2. I believe that, as a future mathematics teacher, I will be able to deliver lessons that are superior to those of other teachers.	3.1	0.55
3. I know effective methods to teach mathematical concepts.	3.1	0.55
4. I am knowledgeable about efficient methods to teach mathematics.	3.2	0.62
5. I understand the process of solving math problems or approaching mathematical concepts.	3.25	0.44
6. I have a good understanding of the mathematical concepts that I need to teach.	3.25	0.55
7. I can use tools such as computers or calculators to instruct the underlying principles of mathematics.	3.2	0.62
8. I possess the skills necessary to teach mathematics.	3.2	0.62
9. I know how to engage students' interest in mathematics.	2.9	0.64
10. I can provide more accurate answers and explanations in response to students' questions	3.3	0.47

pedagogical content knowledge (AI-TPACK) [15], additional practical experience may be needed to achieve higher levels of confidence. This study's qualitative data further illuminates how this growth occurred.

2) QUALITATIVE REFLECTIONS ON TECHNOLOGY USE AND LESSON DESIGN

The reflective journals revealed several common themes in how the pre-service teachers perceived the experience of designing lessons with technology. This study identified three key themes: (1) **Enhanced Problem-Posing, Creativity and Problem-Solving**, (2) **Importance of Pedagogical Context when Technology Tools**, and (3) **Reflection on Teacher's Role vs. Technology Tools' Role**. This study discusses each with example excerpts from participants (using pseudonymous labels S1–S24):

- **Enhanced Problem-Posing, Creativity and Problem-Solving:** Many participants commented that using MATH41 broadened their perspective on crafting problems and activities. With the tool's ability to quickly generate variations, they felt they could be more creative in lesson design. For instance, one participant (S7) wrote, "*MATH41 inspired me to try problem scenarios I wouldn't have attempted otherwise. Seeing how easy it was to modify parameters; I ended up creating a whole set of related problems that explore a concept from multiple angles.*" Similarly, S15 noted that having a swift way to get graphs and computations "*freed up mental energy to focus on the storyline of the lesson and the questions to ask students, rather than getting bogged down in manual problem creation.*" These comments suggest that MATH41 acted as a cognitive and creative amplifier, facilitating pre-service teachers' abilities to devise

more diverse and imaginative lesson scenarios. Moreover, this freedom allowed teachers to engage deeply in meta-cognitive practices, such as reflecting upon and experimenting with different problem conditions and solution strategies known to enhance problem-solving competencies [10], [16]. Thus, through reducing manual workload and providing varied mathematical contexts, MATH41 positively impacted participants' creative and problem-solving pedagogical skills.

- **Importance of Pedagogical Context when Technology Tools:** A recurring caution in the reflections was the need to use Technology Tools thoughtfully within the pedagogical context – echoing what this study expected from the literatures [3] and [4]. Several pre-service teachers recognized that *just because* the tool can generate a lot of content doesn't automatically result in a good lesson. S12 reflected, "*Initially, I was excited by how many problems I could make. But I realized more isn't always better – I had to select or adapt the generated problems to fit my lesson goal and my students' level. I spent a lot of time thinking about which technology tool-generated questions to actually use.*" This quote illustrates a developing understanding that quality and alignment trump quantity. The participants learned to be curators of technology tools outputs, not just consumers. Another participant (S3) wrote that he struggled at first because the automatically created problems "*lacked the specific context I wanted for my class, so I had to tweak the wording and setup.*" He concluded that "*Technology tools can give a great starting point, but the teacher needs to shape it into a meaningful task.*" These reflections reinforce that the *educational context and teacher input are crucial* – a finding aligned with prior studies that warn about unguided technology tools usage [1], [14]. It was encouraging to see participants explicitly noting this; it means the program succeeded in conveying the message that an technology tool is *complementary* to, not a replacement for, principled lesson design. The external teacher reviewers also pointed out this theme as a critical takeaway: one of them commented that the reflections showed a mature understanding that "*technology is helpful only in the hands of a thinking teacher.*"
- **Reflection on Teacher's Role vs. AI' Role:** A deeper theme that emerged, especially in some of the exemplary reflections, was a contemplation of how the teacher's role might change (or not) when using AI in the classroom. Some participants initially feared that relying on AI-generated content might *diminish their role* or creativity as a teacher. By the end, many had the opposite perspective: they saw that it *augmented* their role. For example, S5 wrote, "*Using AI helped me see new ways to present a topic. I still had to decide why and when to use each resource. In fact, I think it made me more creative – I combined the tool's output with my own ideas to make a better lesson than I could have*

unaided.” This sentiment suggests that the experience actually *boosted their confidence* as competent lesson designers who can harness advanced tools. Another reflection by S20 mentioned, “*I learned that an AI tool doesn’t know my students – I do. It’s my job to ensure the content is appropriate. The tool is like an assistant: it can do the heavy lifting, but I have to give it direction and check the work.*” This indicates a clear recognition of the teacher’s irreplaceable role in interpreting and filtering technology tools-generated content for the classroom, aligning with the view that AI can support but not replace teachers [14]. A few participants did acknowledge lingering concerns – for instance, one wondered if over time teachers might become too dependent on such tools for ideas. However, her reflection (S9) concluded optimistically that “*as long as we remain reflective practitioners, we can use these tools without losing our personal teaching touch.*” This highlights the value of the reflective practice itself in ensuring balanced use of technology. Indeed, by engaging in reflection-on-action (after teaching) and reflection-for-action (planning future improvements), the pre-service teachers are developing habits that will help them continually adapt to new technologies in their careers.

Additionally, the course embedded a reflective practice component. After each AI-supported lesson design activity, pre-service teachers completed a brief reflection (via a journal entry or group discussion) examining how AI influenced their design decisions. They documented which AI-generated ideas they adopted or rejected and why. This encouraged them to focus on the learning process and develop metacognitive awareness. Such critical reflection is particularly crucial when using generative AI – it prompted the participants to scrutinize the AI’s suggestions for biases or inaccuracies and to articulate their own rationale in lesson planning. By reflecting on the AI’s role, pre-service teachers cultivated a habit of questioning and contextualizing AI outputs, thereby reinforcing their critical thinking and lifelong learning skills.

Overall, the qualitative data confirms that the AI-integrated teacher education program had a positive impact on the participants’ approach to lesson design. They gained practical skills with a new technology and, more importantly, developed a reflective and critical mindset on how to use that technology in service of learning goals. The mix of enthusiasm for the tool’s possibilities and cautionary understanding of its limits is a promising outcome, suggesting these pre-service teachers are likely to be adaptive and thoughtful in their future technology use. In the next section, this study further interprets these findings and connects them to broader discussions in teacher education and educational technology.

V. DISCUSSION

The findings provide evidence that engaging pre-service teachers in *guided use of AI tools* for lesson design can strengthen their competencies and preparedness for

technology-rich classrooms. In this section, this study discusses the implications of the results in the context of existing literature, the link between lesson design and digital tool use, the role of reflection in teacher learning, and considerations for practice and future research.

A. IMPACT ON LESSON DESIGN COMPETENCY

The high self-assessed competency scores and qualitative feedback indicate that the program was successful in improving participants’ confidence and skills in lesson planning. This extends prior research on technology integration in teacher education. References [2], [3], [4], [6], [10], [15], and [18] argue that because of rapid developments in AI, teacher education must equip pre-service teachers with skills to integrate these technologies, framed by models like TPACK. The study demonstrates a concrete way to do so: by embedding a tool (MATH41) into a methods course and having teachers actively use it in creating lessons. The result was not just tool-specific skill (how to use MATH41) but a broader lesson design competency that included technological, pedagogical, and content considerations – essentially an applied AI-TPACK outcome. Participants learned to *align the use of a digital tool with pedagogical objectives*, which is a critical aspect of lesson design competency in the modern era. For example, deciding which generated problems to use required understanding both the math content and the student audience, reflecting a synthesis of their content knowledge, pedagogy, and technological affordances. This integrated thinking is exactly what frameworks like TPACK encourage, and the results suggest that hands-on experience with AI can concretely foster it. This study saw that participants who might have had only abstract knowledge of “tech integration” before were able to put it into practice meaningfully, echoing calls that pre-service teacher training should include practical AI-related activities to build real competence [1], [2], [4], [15], [17], [19], [31], [32], [33], [34].

B. LINKING LESSON DESIGN AND DIGITAL TOOL USE

A notable observation in the study is how the use of MATH41 influenced the lesson designs produced by the pre-service teachers. The availability of the tool often led to lessons with *richer sets of examples, more visual elements*, and even alternative pathways for students (e.g., an extension activity using extra problems generated by the tool). This suggests a strong linkage: when teachers have powerful digital tools at their disposal *and know how to use them*, they can design more robust learning experiences. This aligns with the notion that technology, when effectively integrated, can enhance instructional design by offering resources and options that would be impractical otherwise [2], [8], [11], [18], [35], [36], [37]. The participants’ lessons serve as small-scale evidence of this – for instance, using an automatically generated graphic allowed a teacher to pose a higher-order question (“What changes if this graph’s equation has a different coefficient?”) that might not have been asked if drawing the graph was a

barrier. In the discussion with participants, some noted that they now see technology as an enabler for differentiating instruction: they could easily create easier or harder versions of tasks to accommodate different learners, a practice recommended for addressing diverse needs. This corresponds to literature pointing out that AI's greatest potential in teaching is in improving adaptivity and personalization [9], [10], [11], [18], [31], [36], [37].

C. VALUE OF REFLECTION IN AI TOOL INTEGRATION

The incorporation of structured reflection activities proved highly valuable in this study. This study found that the reflection journals were the medium through which pre-service teachers solidified their understanding of the role of AI tools in teaching. This reinforces the broader principle that reflection is a critical component of teacher education, especially when new technologies are involved. By reflecting, participants moved beyond the excitement of using a new gadget and critically analyzed its pedagogical value. As [11], [22], and [38] note, creating a habit of reflective practice leads to ongoing improvement in instruction. In this study, reflection enabled participants to identify what worked and what didn't in their AI-assisted lessons, thereby strengthening their future decision-making. The fact that one participant (S9) explicitly mentioned maintaining her "personal teaching touch" through reflection highlights how reflection can guard against the potential pitfall of over-reliance on technology. It encourages *metacognition* – teachers thinking about their own thinking and planning. This study also saw reflection serving as a bridge between theory and practice: participants referenced concepts from their coursework (like differentiation, scaffolding, or student-centered learning) when discussing how they used MATH41, indicating that the reflective process helped them connect the AI tool experience to established pedagogical theory. Therefore, this study suggests that when introducing AI in teacher training, coupling it with systematic reflection (and, if possible, expert feedback) is an effective strategy to ensure deep learning and not just superficial adoption.

D. PEDAGOGICAL IMPLICATIONS

The integration of generative AI tools in teacher education signals a shift in the pedagogical landscape. One key implication is a redefinition of the teacher's role in lesson planning – from sole creator to critical evaluator and co-designer alongside AI. By offloading routine content generation to AI, teachers can devote more effort to higher-order teaching tasks. For example, the AI can supply a draft lesson outline or quiz questions, allowing the teacher to focus on pedagogical refinement, differentiation, and addressing student misconceptions. This resonates with recent arguments that GenAI can support content knowledge acquisition tasks, thereby freeing educators to concentrate on pedagogical modeling, authentic assessment, and fostering students' critical thinking [6], [16], [34], [37].

Consequently, teacher preparation programs should incorporate training not just in using AI, but in scrutinizing and improving AI outputs. This study has, for instance, begun emphasizing AI literacy – including prompt design and bias detection – as core skills for future teachers. Another implication is the democratization of resources: AI-generated lesson materials can serve as open educational resources (OERs), which can reduce dependency on textbooks and expand access to quality materials [16]. This could particularly benefit schools in need, although it also requires that new teachers learn how to adapt AI-provided content to their local context and student needs.

However, an important caveat (and one that the participants understood) is that the *tools must be used with intentional design*. Without clear intent, as prior studies warn, technology can become a distraction or crutch rather than a scaffold [14]. This study highlights that prompt crafting is now a critical teacher skill; the pedagogical outcome can hinge on how the question to the AI is framed. Therefore, a key lesson is that building lesson design competency and digital tool competency together is essential – neither should exist in a vacuum. Pre-service teachers benefitted from seeing directly how their lesson design decisions and tool use decisions interacted.

It was stressed that AI is a support tool, not a replacement for the teacher's own thinking. Throughout the course, pre-service teachers were instructed to use AI-generated outputs critically – for example, checking for accuracy and alignment with curricular goals – rather than copy-pasting content. This safeguard addresses the risk of over-reliance on AI. As recent studies warn, excessive dependence on AI can lead to declines in higher order thinking skills like critical thinking and creativity. In this study, pre-service teachers had to refine and justify each AI suggestion, ensuring they remained actively engaged in the lesson design process. This study explicitly added a note that 'AI-generated lesson plans should be seen as a starting point, with teachers required to evaluate and adapt them to their specific context', echoing the consensus that teachers must not put "all faith" in AI alone [6], [9].

E. COMPARISON WITH PRIOR STUDIES

These findings resonate with and add to several strands of prior research. First, the overall positive impact on lesson design echoes studies that show pre-service teachers can benefit from learning with AI. For example, recent studies by [1] and [12] found that participating in AI-focused courses increased teachers' perception of AI's usefulness and their intention to use it. This study provides a concrete example by detailing the outcomes of such participation. Second, the cautious approach the participants adopted aligns with the results of [14] which suggested that guided use of AI (like their "GPT Tutor" with safeguards) helps maintain learning effectiveness. In the context, the pre-service teachers themselves became the "safeguard" – by deliberately integrating AI outputs into proper lesson structures, they essentially played the role of that guided mediator for their (simulated) students. This bottom-up evidence – teachers figuring out

how to scaffold AI for students – complements the top-down design of AI tutoring systems. Third, This study’s focus on problem generation tools adds to the literature on automated content creation in education. Prior works on automatic math problem generation (whether algorithmic or AI-driven) often emphasize technical quality and student outcomes. This study contributes a teacher-centric perspective: showing that teachers-in-training can effectively use such tools and perceive them as enhancing their capability to design instructions. This is an important angle because ultimately teachers will decide whether and how these tools enter classrooms. If teachers feel empowered by tools like MATH41 (as the data indicates), they are more likely to adopt them, which in turn can influence student experiences.

VI. CONCLUSION

In summary, this study investigated the design and impact of a pre-service teacher education program that integrated a generative AI tool and MATH41 into mathematics lesson planning and instruction. The Introduction of problem generation and visualization tools into the coursework was met with enthusiasm by future teachers and led to tangible improvements in their lesson design outputs. The analyses showed that participants developed the ability to use the tool effectively while keeping pedagogy in the forefront – an outcome evidenced by high self-reported competencies and rich reflective insights. The *Results* indicated that pre-service teachers can leverage AI to create more varied and dynamic lesson content, and perhaps more importantly, they learned to critically evaluate and adapt those AI-generated resources to fit educational needs. Through *Discussion*, this study connected these findings to broader themes such as the necessity of preparing teachers for AI-enhanced education (supporting calls in recent literature for AI readiness in teacher training [16], [18], [31]) and the enduring importance of the teacher’s reflective and design role even as automation becomes available [10].

A. LIMITATIONS

Despite its promising outcomes, this study has limitations. It was conducted in a controlled environment of a single course with a relatively small sample size (24 participants). Given this specific context, results should be interpreted cautiously. Educational settings are heterogeneous, and thus findings from this pre-service mathematics education cohort may not directly apply to other subjects or broader teacher populations (e.g., in-service teachers or other disciplines). Previous literature also emphasizes the difficulty of generalizing outcomes across diverse educational environments [7], [26], [37], [39].

This study also did not measure student learning outcomes since the teaching was done in simulation with peers; it remains to be seen how these AI-integrated lesson design skills translate to real classrooms and impact K-12 students’ learning. The self-assessment nature of the survey means this study rely on teachers’ perceptions of their growth,

which could be inflated by enthusiasm or the desire to please instructors (though the reflections help counterbalance this by revealing struggles and reservations too). Lastly, this study focus was on one particular tool, MATH41, which is quite specific; different tools (e.g., a general chatbot like ChatGPT vs. a domain-specific tool) might yield somewhat different experiences.

Furthermore, this study recognize that not all teacher education programs have ample technological resources. The integration of generative AI, as implemented in this study, presupposed internet access and computing devices, which may be limited in some regions. This raise concerns that GenAI in education could inadvertently widen the digital divide if access is unequal [3], [5], [16].

B. EDUCATIONAL IMPLICATIONS OF INTEGRATING AI TOOLS

Educationally, this study suggests that integrating AI tools into teacher preparation is not only feasible but beneficial. It offers a model for how to blend technical training with pedagogical reflection. The presence of a tool like MATH41 in the curriculum functioned as a catalyst for the pre-service teachers to engage deeply with questions of *how* and *why* to use technology in teaching, rather than just *whether* to use it. This kind of experience is invaluable in the current era, as new teachers will undoubtedly face classrooms where AI and digital tools are commonplace. By having gone through this program, the participants are better equipped to enter that landscape with a mindset of “evidence-informed innovation” – willing to try new approaches, but mindful of student learning principles.

C. HARNESSING AI FOR PEDAGOGICAL INNOVATION

The study contributes optimistic evidence that with thoughtful design, AI tools can be harnessed to enhance teacher education rather than threaten it. The pre-service teachers in this program emerged more competent, confident, and critically aware of how to use technology to enrich mathematics teaching. They exemplify a new generation of educators who see AI as a partner in the creative process of lesson design – one that can handle routine generation tasks and provide inspirations, while the teacher directs the pedagogical vision. As one participant aptly put it, “*the AI can do the heavy lifting, but I hold the compass.*” With their compasses in hand, guided by solid pedagogical training, these future teachers are well-prepared to navigate the evolving landscape of AI in education. This study encourages teacher education programs and researchers to continue exploring this synergy between human pedagogical expertise and artificial intelligence, to ultimately benefit student learning in mathematics and beyond.

D. FUTURE RESEARCH

This study opens several avenues for further inquiry. One direction is to replicate and test the AI-integrated approach with different subject domains (such as science or language

education), larger and varied samples (including in-service teachers), and culturally or resource-diverse contexts. Such studies would provide insight into the broader applicability and any context-specific nuances of these promising results.

Another avenue is to follow these pre-service teachers into their student teaching or first year of teaching to observe if and how they continue to use AI tools and whether their lesson design competency indeed translates to better teaching practice. Longitudinal research could examine retention of these skills and attitudes over time, and test the hypothesized trajectories of long-term impacts, such as embedding reflective practice as a core habit, promoting adaptability to emerging technologies, fostering communities of practice, and shaping a professional identity characterized by data-informed decision making and a growth mindset toward pedagogical innovation.

While this study included in-service teachers as evaluators who reviewed pre-service teachers' lesson plans and reflections, their feedback was primarily used for validating analytical accuracy and identifying exemplary reflections. This study acknowledges the potential to further integrate experienced teachers' perspectives more comprehensively. Specifically, obtaining structured feedback from mentor teachers on the practical relevance of AI-generated lesson plans, real-world applicability in classrooms, educational quality, and the importance of maintaining the human touch could provide valuable insights. Indeed, recent research highlights that experienced teachers are cautiously optimistic about AI tools, recognizing their potential to streamline planning processes while emphasizing the necessity of sustaining educational rigor and the uniquely human dimensions of teaching [1], [6], [9], [31], [36]. Thus, future studies should actively involve in-service teachers in a more systematic feedback process, such as mentoring roles or structured focus groups, to deepen understanding of the practical implications and effectiveness of integrating AI into teaching practice.

Regarding the digital divide and equitable implementation, future research should explore solutions such as establishing shared computer labs or using offline-capable AI tools to ensure all pre-service teachers can participate and recommends future pilots in developing regions to identify best practices for equitable implementation. Targeted training in digital literacy could also assist candidates who are less familiar with such technology.

Finally, the role of reflection could be studied in more depth – for example, comparing guided reflection prompts versus unguided journaling to see how reflection quality influences the integration of technology in lesson design. This study anticipates that the intersection of teacher education and AI will be a fertile ground for research, as the technology and its adoption in schools continue to evolve.

REFERENCES

- [1] R. Kaplan-Rakowski, K. Grotewold, P. Hartwick, and K. Papin, "Generative AI and teachers' perspectives on its implementation in education," *J. Interact. Learn. Res.*, vol. 34, no. 2, pp. 313–338, 2023.
- [2] K. W. Chan, F. Ali, J. Park, K. S. B. Sham, E. Y. T. Tan, F. W. C. Chong, K. Qian, and G. K. Sze, "Automatic item generation in various STEM subjects using large language model prompting," *Comput. Educ., Artif. Intell.*, vol. 8, Jun. 2025, Art. no. 100344.
- [3] M. M. Abualrob, "Innovative teaching: How pre-service teachers use artificial intelligence to teach science to fourth graders," *Contemp. Educ. Technol.*, vol. 17, no. 1, Jan. 2025, Art. no. ep547.
- [4] R. C. Kerr and H. Kim, "From prompts to plans: A case study of pre-service EFL teachers' use of generative AI for lesson planning," *English Teaching*, vol. 80, no. 1, pp. 95–118, Mar. 2025.
- [5] N. S. W. Bo, "OECD digital education outlook 2023: Towards an effective education ecosystem," *Hung. Educ. Res. J.*, vol. 15, no. 2, pp. 284–289, May 2025.
- [6] H. Bae, J. Hur, J. Park, G. W. Choi, and J. Moon, "Pre-service teachers' dual perspectives on generative AI: Benefits, challenges, and integrating into teaching and learning," *Online Learn.*, vol. 28, no. 3, pp. 131–156, Sep. 2024.
- [7] A. Létourneau, M. Deslandes Martineau, P. Charland, J. A. Karran, J. Boasen, and P. M. Léger, "A systematic review of AI-driven intelligent tutoring systems (ITS) in K-12 education," *Npj Sci. Learn.*, vol. 10, no. 1, pp. 1–13, May 2025.
- [8] R. Khazanchi, D. Di Mitri, and H. Drachsler, "The effect of AI-based systems on mathematics achievement in rural context: A quantitative study," *J. Comput. Assist. Learn.*, vol. 41, no. 1, Feb. 2025, Art. no. e13098.
- [9] B. Pepin, N. Buchholtz, and U. Salinas-Fernandez, "A scoping survey of ChatGPT in mathematics education," *Digit. Exper. Math. Educ.*, vol. 2024, pp. 1–33, Sep. 2024.
- [10] W. Wen and H. Wen, "Bridging generative AI technology and teacher education: Understanding preservice teachers' processes of unit design with ChatGPT," *Contemp. Issues Technol. Teacher Educ.*, vol. 24, no. 4, pp. 582–611, 2024.
- [11] D. T. K. Ng, J. K. L. Leung, J. Su, R. C. W. Ng, and S. K. W. Chu, "Teachers' AI digital competencies and twenty-first century skills in the post-pandemic world," *Educ. Technol. Res. Develop.*, vol. 71, no. 1, pp. 137–161, Feb. 2023.
- [12] M. Tight, "Reflection: An assessment and critique of a pervasive trend in higher education," *Eur. J. Higher Educ.*, vol. 14, no. 2, pp. 324–342, Apr. 2024.
- [13] S. J. Oh, "Effective ChatGPT prompts in mathematical problem solving: Focusing on quadratic equations and quadratic functions," *Commun. Math. Educ.*, vol. 37, no. 3, pp. 545–567, 2023.
- [14] H. Bastani, O. Bastani, A. Sungu, H. Ge, Ö. Kabakçı, and R. Mariman, "Generative AI can harm learning," *SSRN J.*, 2024. [Online]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4895486
- [15] I. Runge, F. Hebibi, and R. Lazarides, "Acceptance of pre-service teachers towards artificial intelligence (AI): The role of AI-related teacher training courses and AI-TPACK within the technology acceptance model," *Educ. Sci.*, vol. 15, no. 2, p. 167, Jan. 2025.
- [16] G. van den Berg and E. du Plessis, "ChatGPT and generative AI: Possibilities for its contribution to lesson planning, critical thinking and openness in teacher education," *Educ. Sci.*, vol. 13, no. 10, p. 998, Sep. 2023.
- [17] T. A. D. Gregorio, E. O. Alieto, E.-R. R. Natividad, and M. R. Tanpoco, "Are preservice teachers 'totally packaged'? A quantitative study of pre-service teachers' knowledge and skills to ethically integrate artificial intelligence (AI)-based tools into education," in *Proc. Int. Conf. Digit. Technol. Appl.*, Cham, Switzerland, May 2024, pp. 45–55.
- [18] I. Celik, "Towards intelligent-TPACK: An empirical study on teachers' professional knowledge to ethically integrate artificial intelligence (AI)-based tools into education," *Comput. Hum. Behav.*, vol. 138, Jan. 2023, Art. no. 107468.
- [19] X. An, C. S. Chai, Y. Li, Y. Zhou, X. Shen, C. Zheng, and M. Chen, "Modeling English teachers' behavioral intention to use artificial intelligence in middle schools," *Educ. Inf. Technol.*, vol. 28, no. 5, pp. 5187–5208, May 2023.
- [20] S. Alabidi, A. Owais, F. Alabidi, and O. Taani, "Exploring the role of reflective diaries as a formative assessment strategy in promoting self-regulated learning among ESL students," *Practical Assessment, Res. Eval.*, vol. 27, p. 19, Mar. 2022.
- [21] A. Lama, "Reflective teaching journals as an effective embedded formative assessment process of teaching skill development confidence in a longitudinal medical student-as-teacher elective," *Med. Sci. Educator*, vol. 33, no. 6, pp. 1493–1503, Nov. 2023.

- [22] C. Zhang, J. Schießl, L. Plöbl, F. Hofmann, and M. Gläser-Zikuda, "Evaluating reflective writing in pre-service teachers: The potential of a mixed-methods approach," *Educ. Sci.*, vol. 13, no. 12, p. 1213, Dec. 2023.
- [23] V. D. Wicaksono, B. Budiyo, and S. Supriyono, "Reflective journal development as self-assessment," *EDUTECH, J. Educ. Technol.*, vol. 4, no. 2, pp. 147–153, Dec. 2020.
- [24] A. K. Kim, "Reflective journal assessment: The application of good feedback practice to facilitating self-directed learning," *J. Hospitality, Leisure, Sport Tourism Educ.*, vol. 13, pp. 255–259, Jul. 2013.
- [25] K. Afzali, "Evaluating recall and reflection journals written by pre-service teachers in EFL practicum courses," *Iranian J. Appl. Linguistics*, vol. 21, no. 1, pp. 1–27, Apr. 2018.
- [26] J. König, S. Heine, D. Jäger-Biela, and M. Rothland, "ICT integration in teachers' lesson plans: A scoping review of empirical studies," *Eur. J. Teacher Educ.*, vol. 47, no. 4, pp. 821–849, Aug. 2024.
- [27] C. Antonietti, A. Cattaneo, and F. Amenduni, "Can teachers' digital competence influence technology acceptance in vocational education?" *Comput. Hum. Behav.*, vol. 132, Jul. 2022, Art. no. 107266.
- [28] N. Kang, J. E. Yoo, and H. G. Kim, "Development and validation of a scale of competency of digital age teaching (SCoDAT) for preservice teachers," *Innov. Educ.*, vol. 6, no. 1, pp. 58–77, 2024.
- [29] A. Van Rooyen, "Collaborative exploration of language teachers' digital didactical designs for tablet classrooms," M.S. thesis, Univ. Pretoria, Pretoria, South Africa, 2021.
- [30] M. B. Karsli, K. Meva, K. Sevd, A. Ü. Raziye, and Özge, "Assessment of digital competencies of teacher educators with the DigCompEdu framework," *Int. J. Curriculum Instructional Stud.*, vol. 13, no. 1, pp. 67–94, 2023.
- [31] C. Zhang, J. Schießl, L. Plöbl, F. Hofmann, and M. Gläser-Zikuda, "Acceptance of artificial intelligence among pre-service teachers: A multigroup analysis," *Int. J. Educ. Technol. Higher Educ.*, vol. 20, no. 1, p. 49, Sep. 2023.
- [32] J. Sun, H. Ma, Y. Zeng, D. Han, and Y. Jin, "Promoting the AI teaching competency of K-12 computer science teachers: A TPACK-based professional development approach," *Educ. Inf. Technol.*, vol. 28, no. 2, pp. 1509–1533, Feb. 2023.
- [33] S. Salas-Pilco, K. Xiao, and X. Hu, "Artificial intelligence and learning analytics in teacher education: A systematic review," *Educ. Sci.*, vol. 12, no. 8, p. 569, Aug. 2022.
- [34] M. Nyaaba, "Transforming teacher education in developing countries: The role of generative AI in bridging theory and practice," 2024, *arXiv:2411.10718*.
- [35] O. P. Adelana, M. A. Ayanwale, and I. T. Sanusi, "Exploring pre-service biology teachers' intention to teach genetics using an AI intelligent tutoring-based system," *Cogent Educ.*, vol. 11, no. 1, Dec. 2024, Art. no. 2310976.
- [36] T. Mikeladze, P. C. Meijer, and R. P. Verhoeff, "A comprehensive exploration of artificial intelligence competence frameworks for educators: A critical review," *Eur. J. Edu.*, vol. 59, no. 3, Sep. 2024, Art. no. e12663.
- [37] D. Sotiropoulos and M. Kalogiannakis, "Using GenAI tools to enhance teaching and learning in science," in *Proc. 14th Int. Conf. New Perspect. Sci. Educ.*, Florence, Italy, Mar. 2025, pp. 1–9.
- [38] J. C. Weaver, C. D. Bertelsen, and K. Othman, "Reflection impacts pre-service teachers' instruction and planning," *J. Empowering Teach. Excellence*, vol. 6, no. 2, pp. 3–10, Nov. 2022.
- [39] J. Fischer, M. Bearman, D. Boud, and J. Tai, "How does assessment drive learning? A focus on students' development of evaluative judgement," *Assessment Eval. Higher Educ.*, vol. 49, no. 2, pp. 233–245, Feb. 2024.



SEJUN OH (Member, IEEE) received the B.S. degree in mathematics and the Ph.D. degree in mathematics education from Seoul National University, Seoul, South Korea, in 2009 and 2023, respectively. He is currently an Assistant Professor with the Department of Mathematics Education, Hongik University, Seoul. Previously, he was a High School Teacher at the Ewha Girls' High School and the Keumran High School, affiliated with Ewha Womans University, Seoul. He has published research on developing integrated engineering tools for inquiry-based activity assessment and exploring bio-health applications in mathematics education. His research has been focused on applying artificial intelligence tools in mathematics education, particularly leveraging generative AI for educational research.

...